

TECHNICAL BULLETIN

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**EPOXY CURING AGENT 2080
(ECA 2080)****Typical Properties**

Properties	Value	Test Method
Molecular Weight (average)	246	
Specific gravity @ 25 °C, g/cc	1.00 - 1.03	D-102 (ASTM D 4052)
Appearance	Clear yellow liquid	D-104 (Visual)
Viscosity @ 25 °C, cP	200 - 400	D-150F
Color, Gardner	2 - 4	D-150G
Free acid, %	2 Max.	D-126F

Applications

ECA 2080 is a yellow colored, low viscosity liquid epoxy curing agent based on dodecenylsuccinic anhydride (DDSA). ECA 2080 is designed to cure efficiently in processes such as molding of syntactic foams. This product also performs well as an epoxy curing agent for electrical potting compounds, coatings and conductive adhesives.

ECA 2080 imparts ease of handling due to low viscosity, low toxicity and low volatility. It is a liquid at normal room temperature. If the product crystallizes, it can be returned to liquid by heating 38-55 °C (100-130 °F) for 2-8 hours. When used as an epoxy-curing agent, it has a long pot life at room temperature and gives excellent electrical properties and flexibility to castings.

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For more information on the use of anhydrides like ECA 2080 as epoxy curing agents, please consult the Technical Bulletin, "Formulating Anhydride-Cured Epoxy Systems," available from Dixie Chemical.

Formulation Example

An example formulation is shown in the table below. For this study 125 phr of ECA 2080 was blended with 100 phr of a Standard BPA Liquid Epoxy and 1 phr of 1-methylimidazole (1-MI) catalyst. The sample was cured in the DSC at 10 °C/min to 250 °C. Gel time and glass transition data were obtained:

ECA 2080, phr	BPA Epoxy Resin, phr	1-MI, phr	Gel Time ¹ , min	Tg ² , °C
125	100	1	54	104

1. Sunshine gel time (100 g @ 100 °C)

2. Cured in DSC @ 10 °C /min to 250 °C. Analyzed in DSC: 40 °C /min to 200 °C

Handling & Storage

ECA 2080 will react with water to form diacids. This is normally undesirable, so ECA 2080 should be stored in such a way that it is carefully protected from moisture contamination.

For more details on the design of bulk storage for ECA 2080, consult the Dixie Chemical Company brochure "Epoxy Curing Agent Storage Requirements."

Health Hazards

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Read and understand the relevant Safety Data Sheets (SDS) for all products before use. These anhydrides are primary skin and eye irritants. Avoid contact with skin, eyes, and clothing. Use only with adequate ventilation. In case of contact, follow the procedures outlined in the SDS. Generally, these procedures include immediately flushing the affected skin or eyes with copious amounts of water for at least 15 minutes. In the case of eye contact, get medical attention. Wash contaminated clothing before reuse.

Follow the recommendations in the SDS for personal protective equipment when handling these materials. At a minimum, these procedures typically include protective chemical goggles, impenetrable gloves, and measures to avoid breathing chemical vapors.

Availability

ECA 2080 is available from Dixie Chemical in Pasadena, Texas. Contact your Dixie representative for details.